

Name: _____ Date: _____

Period: _____

Problems 1–6 — Key

1.

Function	Amplitude	Midline
$f(\theta) = 7 \sin(\theta) + 3$	7	$y = 3$
$g(\theta) = -5 \cos(\theta) - 4$	5	$y = -4$
$h(\theta) = \frac{3}{2} \sin(2\theta) + 1$	$3/2$	$y = 1$
$k(\theta) = -\cos(\pi\theta)$	1	$y = 0$

2.

(a) $b = 4$ Period = $2\pi/4 = \pi/2$

(b) $b = \pi/3$ Period = $\frac{2\pi}{\pi/3} = 6$

(c) Period = $\frac{2\pi}{1/4} = 8\pi$

3.

(a) Factor: $2(\theta - \pi/2)$ Phase shift: $\pi/2$ to the right

(b) Factor: $3(\theta + \pi/6)$ Phase shift: $-\pi/6$ (left $\pi/6$ units)

(c) Factor: $\frac{1}{2}(\theta - \pi/2)$ Phase shift: $\pi/2$ to the right

4.

(a) Amplitude: **4** Period: $2\pi/3$ Phase shift: $\pi/6$ right Midline: $y = 5$

(b) Factor: $4(\theta + \pi/4)$

Amplitude: **2** Period: $\pi/2$ Phase shift: $-\pi/4$ (left) Midline: $y = -3$

5. (B) $2\pi/3$ Period = $2\pi/|3| = 2\pi/3$. (The $-\pi$ and $+2$ do not affect the period.)6. (B) $f(\theta) = -6 \sin(\theta) - 1$ Amplitude = $|-6| = 6$. Midline = -1 . (A) has midline $y = 1$; (C) has midline -1 but $\max = 5 \neq 5\ldots$ wait: (B) amplitude 6, midline -1 . Correct.

Teacher Note

Verify all options: (A) midline $y = 1$, no. (B) amplitude 6, midline -1 , yes. (C) $\max = -1 + 6 = 5$, $\min = -1 - 6 = -7$ — midline -1 and amplitude 6, so (C) also works! However (C) has $d = -7$ and the formula shows midline -7 , not -1 . Re-check: $6 \sin(\theta) - 7$ has midline $y = -7$. So (B) is the only correct choice.

Extension optional

Extension (Optional) — Key

(a) Amplitude = $\frac{7 - (-1)}{2} = 4$ Midline $D = \frac{7 + (-1)}{2} = 3$

(b) Period = 4π , so $\frac{2\pi}{B} = 4\pi \Rightarrow B = 1/2$

(c) $p(\theta) = 4 \cos(\frac{1}{2}\theta) + 3$. Using $\cos \theta = \sin(\theta + \pi/2)$: $p(\theta) = 4 \sin(\frac{1}{2}\theta + \frac{\pi}{2}) + 3 = 4 \sin(\frac{1}{2}(\theta + \pi)) + 3$
 $p(\theta) = 4 \sin(\frac{1}{2}(\theta + \pi)) + 3$

Preview — Lesson 3.7: Sinusoidal Function Graphs

You can now read amplitude, period, phase shift, and midline directly from a sinusoidal equation. In Lesson 3.7 we'll go in both directions: from equation to graph (using key points), and from a graph back to the equation. Come ready to use today's parameters to plot and interpret complete sinusoidal graphs.